

Tuned panel + status rail

Directional design advice for the setup panel. Layout, hierarchy, type roles, and state handling. No implementation detail — token names and CSS are yours to choose.

The original panel isn't badly designed; it's under-differentiated. Everything is a small-caps mono label at similar weight, so the eye can't tell a section name from a data source from a system disclosure. Almost every recommendation below is about making the type and containers say what kind of thing you're looking at.

1 • THE CHANGES, RANKED

Ordered by impact per unit of effort. The top four are worth doing even if nothing else is.

01 Give micro-labels four fixed roles, and let no role do two jobs TYPE ONLY

In the original, section headers, field labels, provenance strings, and annotations all render as mono small caps at roughly one weight. That means the reader has to parse text content to learn a label's function, which is exactly the work typography should be doing. Four roles, differentiated on more than one axis each (family, casing, tracking, treatment), and the differentiation holds in grayscale.

02 Make field labels sentence-case sans, not mono caps TYPE ONLY

This is the single highest-leverage move in the list. Once "Speed limit" is sentence case and everything mono-uppercase is either structure or provenance, the panel becomes scannable at a glance — mono caps means "system speaking," sentence case means "your input." It also buys back horizontal space, since caps + tracking is the widest possible way to set a label.

03 Replace the scroll-tracking progress rail with a status rail BEHAVIOR

Because two steps share a row, scroll position can't identify one step — Work never highlights, and the rail reports a position that isn't true. A rail that reports each step's state (complete / needs input / optional—not set) is both correct and more useful: a TCS wants to know what's unresolved, not where the scrollbar is. Clicking still jumps.

04 Move Jurisdiction and Street classification out of the Location column into a full-width band LAYOUT

The density complaint is structural: Location carried a pin summary, corridor list, and two cards while Road carried five short fields, so one column ran roughly twice the other's height. Pulling the two cards down into their own band makes row one balanced and gives each card room to breathe side by side. It also groups the two confirm/dismiss flows together, which is where a reviewer looks for them.

05 Split the jurisdiction block into two single-purpose cards GROUPING

The original card mixed three text sizes and two independent suggestion flows under one header, so there was no way to tell which buttons applied to which value. One card, one decision, one proposal. Same total pixels, half the ambiguity.

06 Make the suggestion row one repeated shape, wherever it appears COMPONENT

Both suggestions do the same thing — propose a value with evidence and wait — but they read as two different components. One shape (glyph, "X suggests," the value, confirm, dismiss, evidence line underneath) teaches the pattern once. Two explicit buttons stay, and nothing auto-applies.

07 Show what a confirm or dismiss did, using the disclosure shape BEHAVIOR

In the original, resolving a suggestion presumably just removes the row, which erases the record of a decision an inspector may need to see. Replace the resolved proposal in place with the same block shape carrying ✓ or ×, its evidence, and an undo. The disclosure block becomes the general shape for "the system or you changed something," not a one-off warning.

08 Give the disclosure note a heavier container, not a lighter one EMPHASIS

It is the most consequential sentence on the screen — the system changed a value the user didn't change. It gets the amber rule, a border, a full-size glyph, and a full sentence with the old and new values bolded, plus its provenance on a second line in the provenance role so it doesn't compete with the sentence.

09 Set the corridor extent as a table with tabular figures, right-aligned, unit demoted DATA

Five values that get compared to each other should form a column you can read down. Right-aligned tabular numerals with "ft" set smaller and dimmer makes the magnitudes line up, which is how a reviewer checks them. Every row and value stays; the block gets shorter mainly by tightening leading.

10 Add a proportional bar under the extent table DATA

The numbers are the legal record; the bar is the sanity check. It's the fastest way to catch a work zone that dwarfs its advance warning, and it costs about nine pixels of height. Segment order matches the table order, so it reads as a summary of the rows above rather than new information.

11 Fill Schedule's empty side with the jurisdiction-window rows in an unevaluated state EMPTY STATE

The dead half-panel at "Not set" isn't a layout hole to fill with decoration — it's a missing answer to "what happens when I do set dates." Show the same four rows, same order, same widths, with a circle glyph and "— set dates to check." When dates arrive, only the glyph and value change, so the user's eye doesn't have to re-find anything.

12 **Mark provenance with a dotted underline, not just dim color** TYPE ONLY

Provenance is the one label role that's also an affordance — an inspector may want to know where the data came from. A dotted underline reads as "inspectable" and survives grayscale, where a dim gray alone is just small text. It also keeps provenance visually distinct from field labels at the same size.

13 **Set a detected-vs-applied reference block in the Road column** GROUPING

Detection source, detected value, and applied value are three facts of the same kind, currently scattered as annotations next to unrelated fields. As a three-row block they answer the inspector's question in one place, and they give the sparse Road column legitimate mass instead of filler.

14 **Promote the pin readout to a fact strip** LAYOUT

Coordinates, bearing, and speed are read, not edited, so they shouldn't look like fields. A bordered strip of labeled cells reads as instrument output in one glance and separates cleanly from the editable stack below it.

15 **Use one glyph vocabulary across the whole panel** ACCESSIBILITY

! changed · ~ proposed · ✓ confirmed or passing · × dismissed · ○ unevaluated · — optional. This is what makes the panel grayscale-safe in practice rather than in principle: every colored thing has a glyph carrying the same meaning, and the glyph set is small enough to learn passively.

Label roles, as designed

ROLE	TREATMENT	JOB — AND ONLY THIS JOB
Section	mono 500 · 10px · 0.20em UPPERCASE · brightest ink	Names a section or card. One per container, always on its top edge, never inside a field.
Step index	mono 400 · 10px · 0.14em UPPERCASE · dim	The step number and nothing else. Right edge of the section header; the rail's jump target.
Field label	sans 500 · 12px · 0 tracking Sentence case · mid	Names one input. Sentence case is the tell: if it labels a control, it is never uppercase.
Provenance	mono 400 · 10px · 0.04em lowercase · dim · dotted underline	Where a value came from, and code citations. The underline marks it inspectable.

Disclosure text is deliberately **not** a label role. It's body copy in a container — amber rule, glyph, full sentence — sitting above all four label tiers in the hierarchy.

2 · THE PRINCIPLES

One role per treatment, differentiated on two axes.
Any two label roles differ in at least two of family, casing, size, tracking, color, and decoration. One axis of difference (dimmer gray) fails in print, in grayscale, on a sun-washed tablet in a truck cab, and for anyone who hasn't memorized your palette. This is the rule that fixes the small-caps overload, and it's the one to apply to screens I haven't seen.

Mono caps means the system is speaking. Sentence-case sans means the user is.

Once that split is consistent, a user can tell input from output without reading a single word. It's worth protecting even when a mono label would fit a tight space better.

A value the user didn't set is a system event, not a field annotation.

Test: if the system chose, changed, inferred, or proposed the value, it needs a container — a bordered block with a glyph and a full sentence naming the old and new values. If the user set it, an inline provenance line is enough. Annotations get scanned past; containers don't.

State changes reuse the container they replace.

Confirmed, dismissed, and changed all render in the same block shape at the same place in the flow, differing by glyph, rule color, and sentence. Nothing vanishes on resolution, so the panel keeps a visible record of who decided what — and users learn one shape instead of four.

Navigation derives state; it never stores its own.

The rail computes each step's status from the actual field values on every change, rather than tracking scroll or recording "visited." Derived status can't drift out of sync with the form, and it answers the more useful question. Corollary: if the rail can't derive an honest answer, it should say less rather than guess — dismissing a suggestion without picking a value leaves the step at "needs input," not a false ✓.

Empty states show the shape of the answer.

An unresolved region renders its real rows with an unevaluated glyph and an em-dash value, at final widths and in final order. Filling dead space is a side effect; the point is that the user learns what the field will tell them, and that nothing reflows when the data lands.

Density is a property of a region, not of a column.

Two columns forced to share a row will always fight, because content volume per step isn't equal. Let a step take the width it needs, and pair steps only when their natural heights are comparable.

Color is the second signal, always.

Design every state in grayscale first, then add color. If a state reads correctly without color, color is doing what it should — accelerating recognition rather than carrying meaning.

3 • WHAT I DELIBERATELY KEPT

- **Two explicit buttons on every suggestion.** No single-tap apply, no undo-toast pattern, no auto-apply with a disclosure. Confirming a jurisdiction has permitting consequences; the friction is the feature.
 - **Every corridor row and footage, spelled out.** The bar is additive. I compressed leading and demoted the unit, but no row was merged, abbreviated, or moved behind a toggle.
 - **All provenance strings.** OSM DETECTION, US Census TIGER, MEDIAN PRESENT and the full TIGER vintage line all survive, restyled to one voice and moved next to the value they explain. The TIGER caveat about confirming with the permitting authority stays in full — it's the most legally loaded sentence in the panel.
 - **The monospace-and-dark engineering aesthetic.** It reads as instrument, not consumer app, which is right for this audience. I didn't soften corners, add elevation, or warm the palette.
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- **Numbered steps as visible structure.** Still labeled STEP 2 ... STEP 5, still the rail's jump targets. The rail changed what it reports, not what it points at.
- **Schedule as genuinely optional.** It reads "optional · not set" and stays neutral rather than nagging. An unset optional step is not an error.
- **Square corners, hairline rules, no shadows.** Containers are separated by 1px rules and background shifts. Consistent with the existing generator surfaces.

4 • RISKS AND TRADE-OFFS

The full-width jurisdiction band costs vertical height.

Row one gets balanced, but the panel gets taller and the two cards move further from the pin they relate to. If your users work mostly on a laptop and scroll a lot, weigh this against just capping the Location column's height.

Sentence-case field labels will look "less technical" to some of your users.

Crews accustomed to the all-caps readout may read it as softer. I think the scanning gain wins decisively, but it's the change most likely to draw a reaction, so land it with the rest of the type system rather than alone.

Persistent resolved-state blocks accumulate.

With two suggestions it's fine. With eight, the panel becomes a log. If suggestion volume grows, the resolved blocks should collapse into a per-section summary line that expands — which is a lighter version of the ledger direction, and worth revisiting then.

Below roughly 900px the two-column rows must stack.

Stacked, the panel is long and the status rail becomes the primary navigation, so its accuracy matters more, not less. I did not design a tablet or phone layout; if crews use this in the field, that's the next thing to design and it may change the band decision.

The corridor bar can mislead at extremes.

A 50 ft taper next to a 3,000 ft work zone renders as a sliver, so the bar needs a minimum segment width and shouldn't be trusted for reading small values — the table is the source of truth. Decide whether to label segments inside the bar or leave it unlabeled; I left it unlabeled to keep the table authoritative.

States I couldn't design.

Validation and conflict errors (a work zone longer than the corridor, a speed the jurisdiction disallows), loading and failure states for detection, a jurisdiction with no window data on file, and multi-jurisdiction pins on a boundary. The last one matters most: the TIGER caveat implies it happens, and a single-value suggestion row can't express it.

GUESS I MADE • RAIL STATUS CATEGORIES

I assumed exactly three step states — complete, needs input, optional-and-unset — and that Steps 3 and 4 are always complete because they have defaults. If a step can be complete-but-stale (detection changed after the user edited), or partially complete, the rail needs a fourth state and I'd want to design its glyph rather than have one picked ad hoc.

GUESS I MADE • WHAT MAKES LOCATION "COMPLETE"

I treated "None — baseline" as a real, valid answer for jurisdiction, and an unset street classification as blocking. That's a product rule, not a design one. If baseline should nag until explicitly confirmed, the rail's logic changes, not its design.

GUESS I MADE • THE SCHEDULE WINDOWS

Peak-hour restriction, night work window, permit lead time, and holiday blackout are my invention of four plausible rows. The design assumes a fixed row set per jurisdiction that stays stable between the unset and evaluated states. If jurisdictions expose different window types, the unevaluated state can't show real rows and needs a different empty state.

GUESS I MADE • UNDO ON A CONFIRMED SUGGESTION

I gave confirmed and dismissed blocks an undo. If confirming a jurisdiction triggers downstream permit logic, undo may not be cheap, and the affordance should go.

5 • VISUAL REFERENCE

Directional, not pixel truth. Spacing, exact type sizes, and the palette are yours; what these images carry is hierarchy, grouping, and state behavior. Both captures come from the working prototype — the select and date inputs render blank in capture, which is a screenshot artifact.

REFERENCE A – PENDING SUGGESTIONS, ONE CHANGED VALUE, SCHEDULE NOT SET

02 LOCATION

2 TO CONFIRM

03 ROAD

SET

04 WORK

SET

05 SCHEDULE

OPTIONAL · NOT SET

LOCATION

STEP 2

COORDINATES

39.7401, -104.9827

BEARING

90°

SPEED

30 mph

CLASSIFICATION

Urban arterial · divided

LANES

2 per direction

!

Road type set to **Urban arterial**, changed from detected **Rural — divided**.

applied from picker · osm detection

CORRIDOR EXTENT

1,600 ft

—

Advance warning

300 ft

--

Taper

50 ft

...

Buffer

200 ft

—

Work zone

1,000 ft

...

Downstream

50 ft

EDIT LOCATION & CORRIDOR →

ROAD

STEP 3

Road type

Urban arterial

osm detection · rural — divided

Speed limit

30 mph

mutcd: <45 mph uses $L=W\cdot S^2/60$

Lanes per direction

12 ft

1

2

3

4

Lane width

12 ft

☒ Divided highway

median present · osm

DETECTED

Rural — divided

APPLIED

Urban arterial

SOURCE

OSM detection · 2025

JURISDICTION & CLASSIFICATION

STEP 2 · CONTINUED

JURISDICTION

None — baseline (MUTCD + CDOT only)

Statewide baseline — MUTCD + Colorado Supplement only.

PIN SUGGESTS

Denver

Confirm

Dismiss

pin inside denver municipal limits · us census tiger/line place boundaries, 2025 · approximate — confirm with permitting authority

STREET CLASSIFICATION

Local

Collector

Arterial

DETECTED ROAD SUGGESTS

Arterial

Confirm

Dismiss

osm primary

WORK

STEP 4

Work type

Utility locate

Work zone length (ft)

1000

☐ Night operation

+ retroreflective

☐ Apply work-zone speed reduction

lower limit through zone

SCHEDULE

STEP 5

Work dates

Single day

Date range

Not set

JURISDICTION WINDOWS

reference only

o Peak-hour restriction

— set dates to check

o Night work window

— set dates to check

o Permit lead time

— set dates to check

o Holiday / event blackout

— set dates to check

windows resolve against the selected jurisdiction once dates exist

Status rail shows ● 02 Location · 2 to confirm, ✓ on Road and Work, — optional on Schedule. **Amber disclosure** sits directly

under the Location facts, heaviest container in the panel. **Both suggestion rows** use the same shape in the Step 2 continued band, evidence line beneath each. **Schedule's right side** carries four ○ unevaluated rows rather than empty space.

REFERENCE B – ONE CONFIRMED, ONE DISMISSED, DATES SET

02 LOCATION

CLASSIFICATION NOT SET

03 ROAD

SET

04 WORK

SET

05 SCHEDULE

4 DAYS

LOCATION

STEP 2

COORDINATES

39.7401, -104.9827

BEARING

90°

SPEED

30 mph

CLASSIFICATION

Urban arterial · divided

LANES

2 per direction

! Road type set to Urban arterial, changed from detected Rural — divided.

applied from picker · osm detection

CORRIDOR EXTENT

1,600 ft

— Advance warning

300 ft

-- Taper

50 ft

... Buffer

200 ft

— Work zone

1,000 ft

... Downstream

50 ft

EDIT LOCATION & CORRIDOR →

ROAD

STEP 3

Road type

Urban arterial

osm detection · rural — divided

Speed limit

30 mph

mutcd: <45 mph uses $L=W\cdot S^2/60$

Lanes per direction

12 ft

12 ft

☒ Divided highway

median present · osm

DETECTED

Rural — divided

APPLIED

Urban arterial

SOURCE

OSM detection · 2025

JURISDICTION & CLASSIFICATION

STEP 2 · CONTINUED

JURISDICTION

None — baseline (MUTCD + CDOT only)

Denver — MUTCD + CDOT + Denver Right-of-Way rules.

✓ Jurisdiction set to Denver — you confirmed the pin suggestion.

us census tiger/line place boundaries, 2025

UNDO

STREET CLASSIFICATION

Local

Collector

Arterial

✕ Detection Arterial dismissed. Classification left unset.

osm primary

RESTORE SUGGESTION

WORK

STEP 4

Work type

Utility locate

Work zone length (ft)

1000

☐ Night operation

+ retroreflective

☐ Apply work-zone speed reduction

lower limit through zone

SCHEDULE

STEP 5

Work dates

Single day

Date range

Not set

Start

End

DENVER WINDOWS

checked against 4 days

! Peak-hour restriction

no closure 07-09, 16-18

✓ Night work window

22:00-05:00 permitted

! Permit lead time

5 business days — 2 left

✓ Holiday / event blackout

none in range

windows resolve against Denver · re-checked on every date change

Confirmed and dismissed reuse the disclosure container with ✓ and × and keep their evidence plus an undo — the decision stays on the record. **The rail** still reads "classification not set" after the dismiss, because it derives from the value, not from whether the row is gone. **Schedule windows** resolved against Denver: same four rows, same widths, only glyph and value changed.

REFERENCE C — THE SCHEDULE BLOCK IN BOTH STATES, SIDE BY SIDE

SCHEDULE REFERENCE BLOCK

BOTH STATES

state 1 · not set

JURISDICTION WINDOWS	reference only
○ Peak-hour restriction	— set dates to check
○ Night work window	— set dates to check
○ Permit lead time	— set dates to check
○ Holiday / event blackout	— set dates to check

Rows are present and readable before dates exist, so the panel never shows an empty half. Circle glyph = unevaluated.

state 2 · sep 8 - sep 11, denver

DENVER WINDOWS	checked against 4 days
! Peak-hour restriction	no closure 07-09, 16-18
✓ Night work window	22:00-05:00 permitted
! Permit lead time	5 business days - 2 left
✓ Holiday / event blackout	none in range

Same four rows, same order, same widths — only glyph and value change. ! and ✓ carry the verdict without color.

The empty state is the evaluated state with its verdicts removed. ○ unevaluated, ! restriction applies, ✓ clear — verdict legible with no color at all.